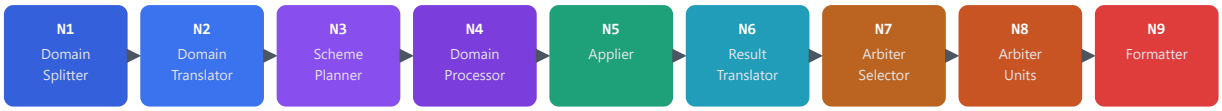
Orchestrator Experiment Report

T34 Phase Transition & P3 Learning From Failure · 2026-05-04 · Project UmFromScratch

PHASE 24 — T34 | PHASE 25 — P3 | 1014 TESTS PASSING | PASS_WITH_GAPS | BLOCK → LEARN → VERIFY

1. The System



22-PREDICATE VOCABULARY — FIXED INTER-UNIT CONTRACT



PHASE CONTEXT PACKET

Structured artifact produced at the end of each phase. Carries decisions, constraints, gaps, and markers into the next phase.



2. T34 — Phase 3 → Phase 4 Transition Gate

WHAT T34 PROVES

For the first time, the system verifies that a Phase 4 (Implementation) source packet is consistent with what Phase 3 (Technical Design) decided. Three scenarios are evaluated through the Transition Gate.

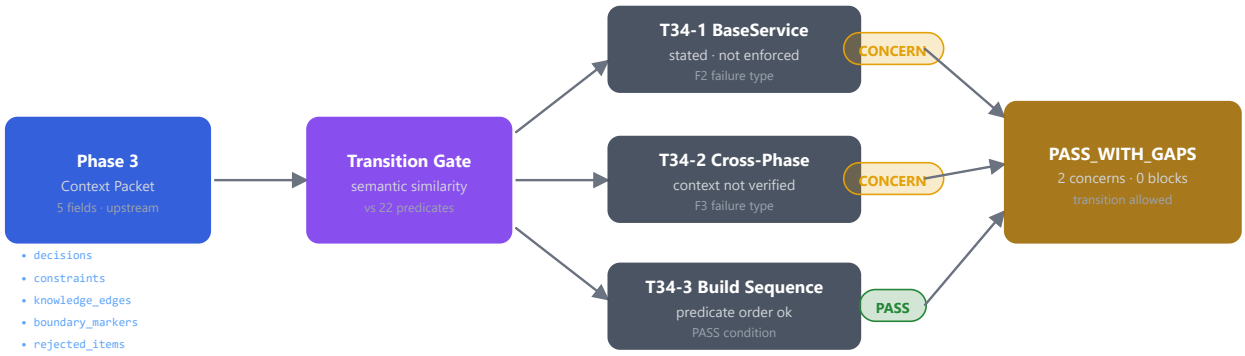
10 domain packets | 3 scenarios | 2 concerns | 0 blocks

NEW CODE — T34PHASETRANSITIONPROOF.CS

RECORD TYPE	KEY FIELDS
`PhaseContextPacket`	decisions · constraints · knowledge_edges · boundary_markers · rejected_items

`DomainPacket`	packetId · description · predicates[] · status
`TransitionScenario`	scenarioId · reason · verdictType · blocksTransition · evidencePacketIds[]
`TransitionGateResult`	overallVerdict · scenarios[] · lostConstraints · concernCount
`TransitionEvidence`	verdict · timestamp · runId · phaseFrom · phaseTo
`Metrics`	17 named metrics · experiment_success JSON key

TRANSITION GATE FLOW — T34 RUN



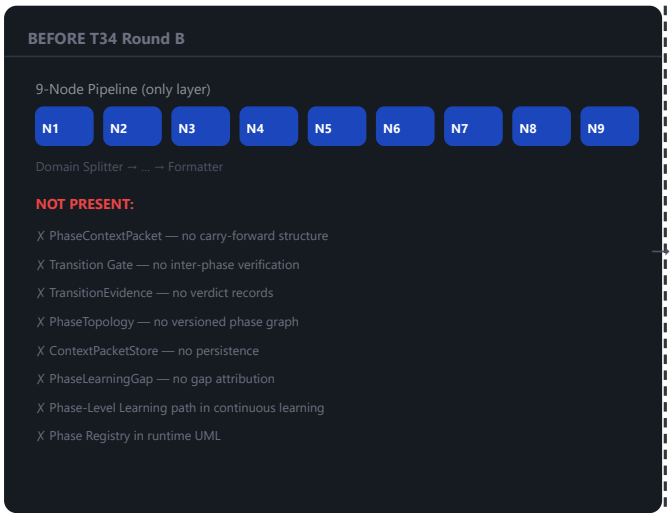
10 DOMAIN PACKETS — PHASE 4 SOURCE SET

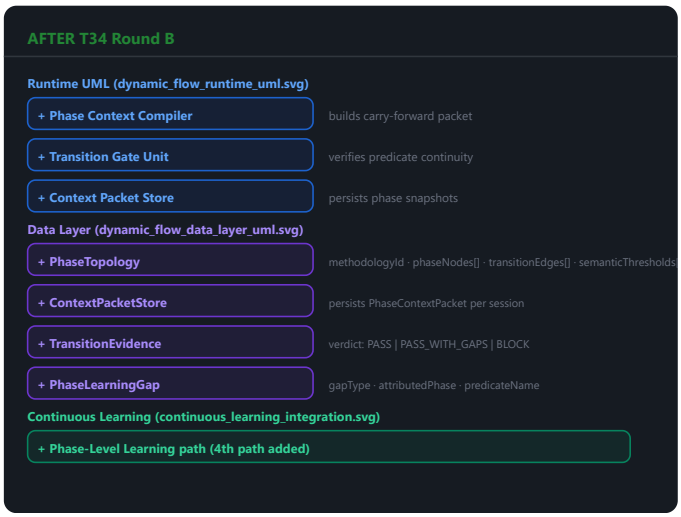
ID	DESCRIPTION	STATUS	SCENARIO
td01	Runtime orchestration pattern decision	present	T34-1 evidence
td02	Service separation rule	present	T34-1 evidence
td03	Memory allocation pattern	present	baseline
td04	Error propagation rule	present	baseline
td05	Predicate vocabulary boundary	present	baseline
td06	Base service inheritance constraint	concern	T34-1 · **P3-C removes this**
td07	Cross-phase context propagation	concern	T34-2 evidence
td08	Build-time type checking	present	T34-3 evidence

td09	Interface segregation boundary	present	baseline
td10	Dependency injection scope rule	present	baseline

3. Architecture Changes — UML Before / After

FOUR UML DIAGRAMS UPDATED DURING T34 ROUND B





MISSING_TOOLS_FOR_CONTEXT_COMPLETION.SVG

Moved entities from the *Designed/Not Built* bucket into *Exists/Proven*: Phase Context Compiler, Transition Gate Unit, Context Packet Store.

The *Missing/Now Required* section names the next layer: Phase Registry with versioned topology.

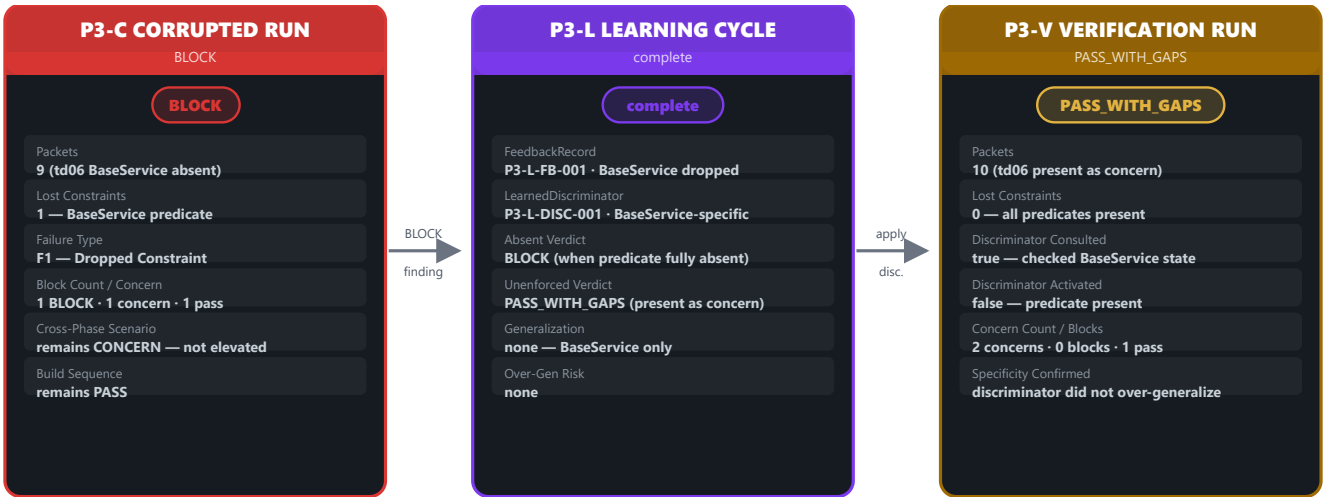
CONTINUOUS_LEARNING_INTEGRATION.SVG

Added a **4th learning path**: Phase-Level Learning, alongside Real-Time Request, Idle Scheduled, and Gap-Driven Cell Creation paths.

Phase-Level Learning fires after a transition gate run and writes gap packets attributed to the originating phase.

4. P3 — Learning From Transition Failure

P3 THREE-STAGE LEARNING CYCLE



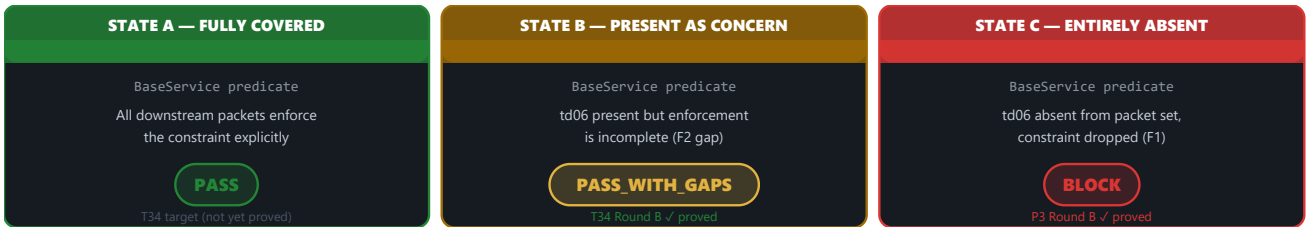
NEW RECORD TYPES — P3LEARNINGFROMFAILUREPROOF.CS

RECORD	PURPOSE
`FeedbackRecord`	Captures a BLOCK finding with named dropped predicate
`LearnedDiscriminator`	Encodes how absent vs. present-as-concern differ
`LearningCycleResult`	Wraps feedback + discriminator with over-gen risk
`CorruptedRunResult`	P3-C output — 9-packet BLOCK run
`VerificationRunResult`	P3-V output — 10-packet re-run with discriminator check

8 PASS GATES — ALL GREEN

GATE	CONDITION	RESULT
P3-1	P3-C verdict = BLOCK	✓
P3-2	LostConstraintCount = 1, BaseService	✓
P3-3	Cross-phase stays concern not block	✓
P3-4	FeedbackRecordsAdded = 1	✓
P3-5	DiscriminatorsAdded = 1, no gen.	✓
P3-6	P3-V verdict = PASS_WITH_GAPS	✓
P3-7	ConcernCount = 2, BlockCount = 0	✓
P3-8	DiscriminatorActivated = false	✓

BASESERVICE PREDICATE — THREE PROVABLE STATES



5. Results & Next Steps

1014	GAPS	BLOCK	GAPS	8/8
TESTS PASSING	T34 VERDICT	P3-C VERDICT	P3-V VERDICT	GATES PASSED

NONE	+17	2
OVER-GEN RISK	TEST Δ	COMMITS

WHAT COMES NEXT — P4

Prove that a topology of phases orchestrates a full lifecycle: Phase 1 (UI/UX) → Phase 3 (Technical Design) → Phase 4 (Implementation).

Pass conditions: each phase produces a carry-forward packet, the next phase consumes it, both transition verdicts are correct, gaps are attributed to the originating phase.

GATE B CLOSURE — BASESERVICE

T34 flagged BaseService as a concern. The next concrete action is adding a Phase 4 Gate B assertion that explicitly enforces the constraint.

Once Gate B passes, State A (fully covered → PASS) will be proved and all three discriminator states will have evidence.

Appendix — Artifact Locations

ARTIFACT	LOCATION	COMMIT
T34 Oracle Documents (7 files)	`orchestrator experiment/T34-PHASE-TRANSITION-ROUND-A-2026-05-04/`	`e6622a8`
T34 C# Experiment	`35.ExperimentsV2/Experiments/T34PhaseTransitionProof.cs`	`e6622a8`
T34 Tests (8 tests)	`Tests/DynamicFlowT34Tests.cs`	`e6622a8`
P3 Oracle Documents (6 files)	`orchestrator experiment/P3-LEARNING-FROM-FAILURE-ROUND-A-2026-05-04/`	`8e4134b`
P3 C# Experiment	`35.ExperimentsV2/Experiments/P3LearningFromFailureProof.cs`	`8e4134b`
P3 Tests (9 tests)	`Tests/DynamicFlowP3Tests.cs`	`8e4134b`
Architecture SVGs (4 files)	`orchestrator experiment/dynamic_flow_*.svg`	—
This Report (HTML + PDF)	`orchestrator experiment/EXPERIMENT-REPORT-T34-AND-P3-2026-05-04-VISUAL.*`	—